exerts controlling torque. The two coils are coiled in opposite directions.

The inner ends of both the hair springs are attached to the spindle. The outer end of spring A is attached to lever which is actuated by a set screw mounted at the front of the instrument. So zero setting can be easily done without removing the cover. The outer end of spring B is fixed.

When the pointer moves under the influence of deflecting torque, one of the spring unwind itself while the other gets twisted. The twist produces the controlling torque which is directly proportional to the angular deflection of the pointer.

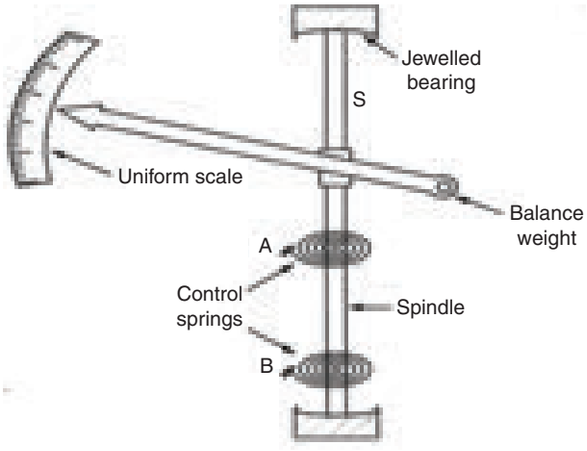


Fig. 6.1 Spring control

ii. Gravity control

The arrangement of gravity control system is shown in figure 6.2, in which two adjustable weights, control weight and balance weight are attached to the spindle. The control weight provides a controlling torque due to gravity while the balance weight is used for balancing the weight of the moving system.

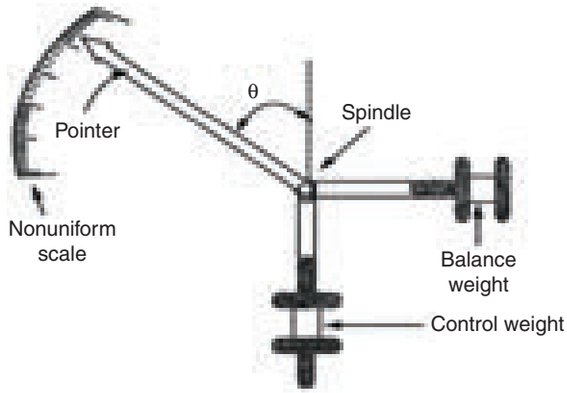


Fig. 6.2 Gravity control

6.3.3 Damping torque

The deflecting torque provides some deflection and controlling torque acts in the opposite direction to that of deflecting torque. So before coming to the rest, pointer always oscillates due to inertia, about the equilibrium position. So to bring the pointer to rest within short time, damping torque is required.

If the moving system reaches to its final position rapidly but smoothly without oscillations, the instrument is said to be critically damped. If the damping torque is more than what is required for critical damping, the instrument is said to be over damped. In over damped case the response of the system is very slow and lethargic.

If the instrument is under damped, the moving system will oscillate about the final steady position with decreasing amplitude and will take some time before it comes to rest.

The following three methods are used for producing damping torque.

i. Air friction damping
ii. Fluid friction damping
iii. Eddy current damping

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